



ABAP Platform in SAP S/4HANA Cloud Private Edition and SAP S/4HANA 2025

Overview and product highlights

Carine Tchoutouo Djomo, Olga Dolinskaja, SAP
23. June 2026

AGENDA

ABAP PLATFORM OVERVIEW

ABAP PLATFORM FOR SAP S/4HANA EXTENSIBILITY

THE ABAP CLOUD DEVELOPMENT MODEL

ABAP PLATFORM 2025 PRODUCT HIGHLIGHTS

- Generative AI in ABAP
- Clean core development
- ABAP development tools for Eclipse
- ABAP language
- Database-centric optimizations
- ABAP RESTful application programming model
- Embedded analytics
- Integration services

ADDITIONAL RESOURCES



ABAP Platform 2025 - Overview and Product Highlights



Scan to access the blog post (incl. full slide deck) or click on the link:
[ABAP Platform in SAP S/4HANA Cloud Private Edition and SAP S/4HANA 2025](#)

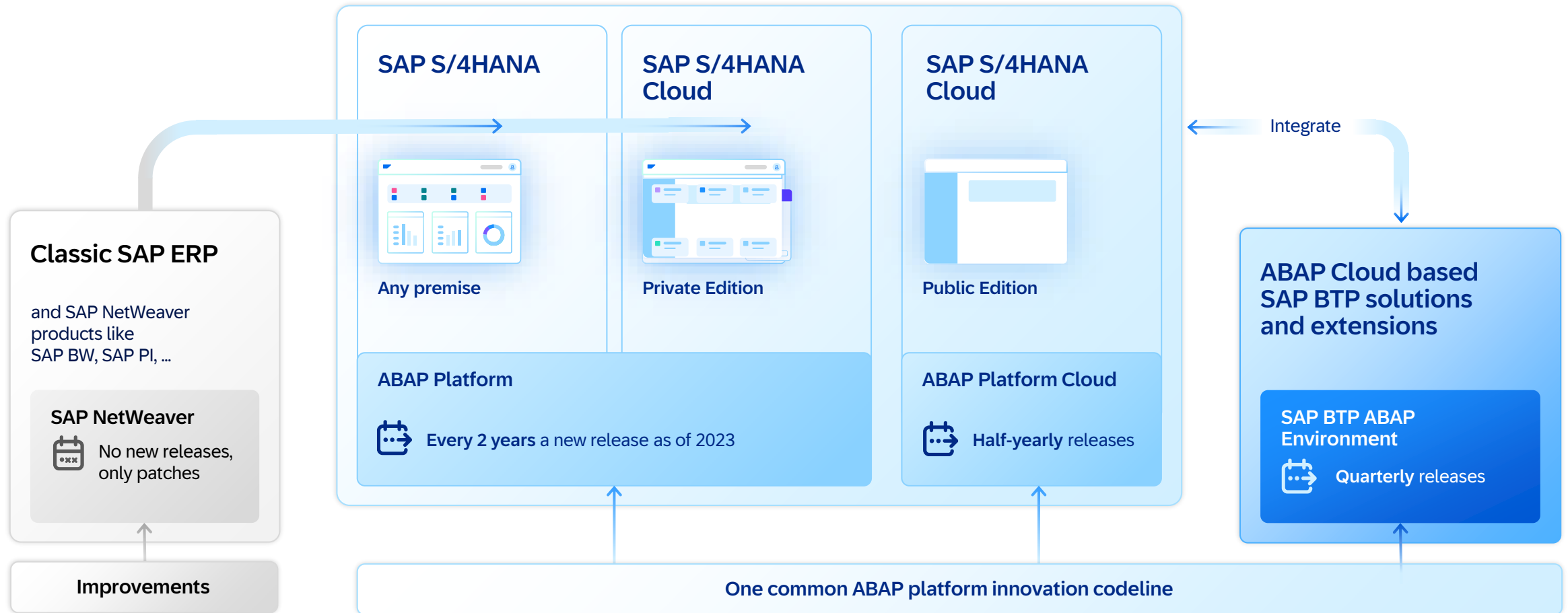


ABAP PLATFORM

OVERVIEW



ABAP as main technology provider for SAP S/4HANA



ABAP Platform 2025 at a glance



ABAP Cloud development model to build cloud-ready business apps, services, and extensions that adhere to clean core principles by definition, while coexisting with classic ABAP development

SAP Joule for Developers, ABAP AI capabilities to accelerate ABAP development and transform custom code from ECC to SAP S/4HANA, and the **ABAP AI SDK** to infuse AI into custom development

Refined SAP S/4HANA extensibility model based on **clean core level concept** using ABAP Cloud as default, and enabling classic ABAP for clean core if following SAP guidelines for extensibility governed by ABAP test cockpit clean core checks

Boost development process with new features and productivity improvements in the **ABAP language** and the **ABAP development tools for Eclipse**, and now also with the **ABAP development tools for VS Code**, including an agentic AI developer experience

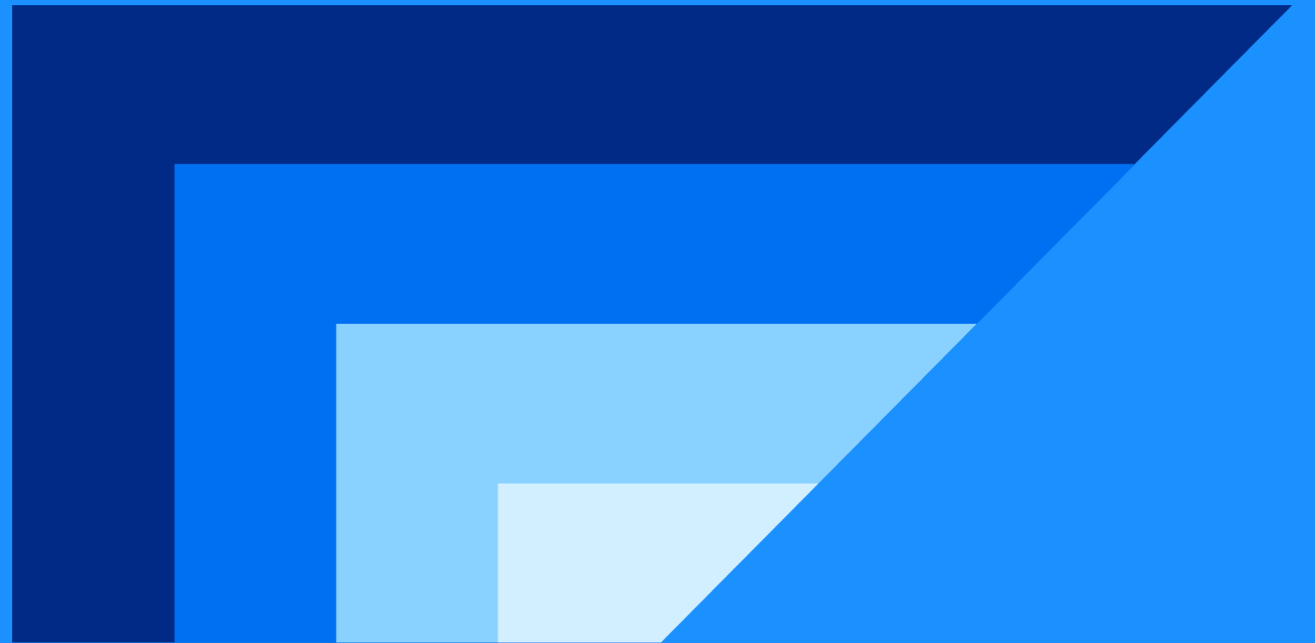
New advanced, database-centric features **enriching data integration scenarios**. New write scenarios, use of CDS external entities, and consumption of imported CDS models make it easier than ever to integrate external data into ABAP CDS view stack

New features to accelerate and enhance the development of transactional SAP Fiori apps and APIs with the **ABAP RESTful Application Programming Model**, including new RAP generators, cross-BO scenarios, event-driven side effects, collaborative draft, treeviews, end-to-end extensibility options

Facilitated development and management of **embedded ABAP analytics** scenarios by using ABAP Cloud generators for multidimensional analysis apps and Review Booklet apps, and by using reusable measures and filters

Strengthened event-driven architecture of SAP S/4HANA and improved interoperability across enterprise solutions through **seamless integration with SAP Event Mesh Advanced Plan** and **support of SAP Cloud Application Event Hub** to consume events published by SAP Order Management Foundation on SAP BTP

ABAP PLATFORM FOR SAP S/4HANA EXTENSIBILITY



Extensibility portfolio for SAP S/4HANA Cloud¹

ABAP Cloud in SAP Build

/ More information in the [clean core extensibility deck](#)

Use SAP Build to extend
the way you need it:

Customized standard apps
with ABAP Cloud
and SAPUI5 flexibility

Custom apps and APIs
with ABAP Cloud, SAPUI5,
and SAP Fiori

SAP S/4HANA Cloud
**On-stack
extensibility**

Streamlined
integration

SAP BTP
**Side-by-side
extensibility**

Custom apps and APIs
with ABAP Cloud, SAP Cloud
Application Programming Model,
SAPUI5, and SAP Fiori

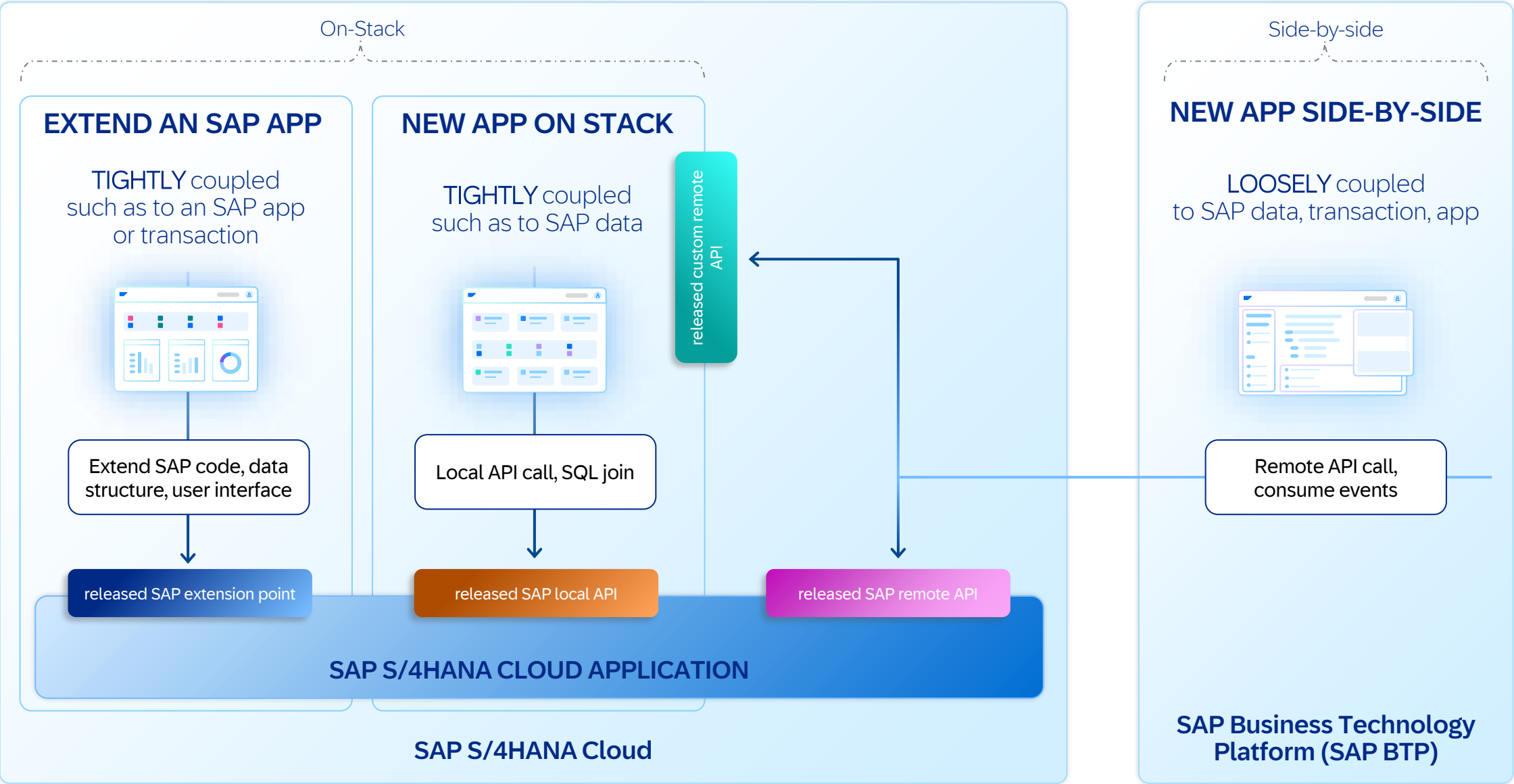
Integrations

Tasks and processes

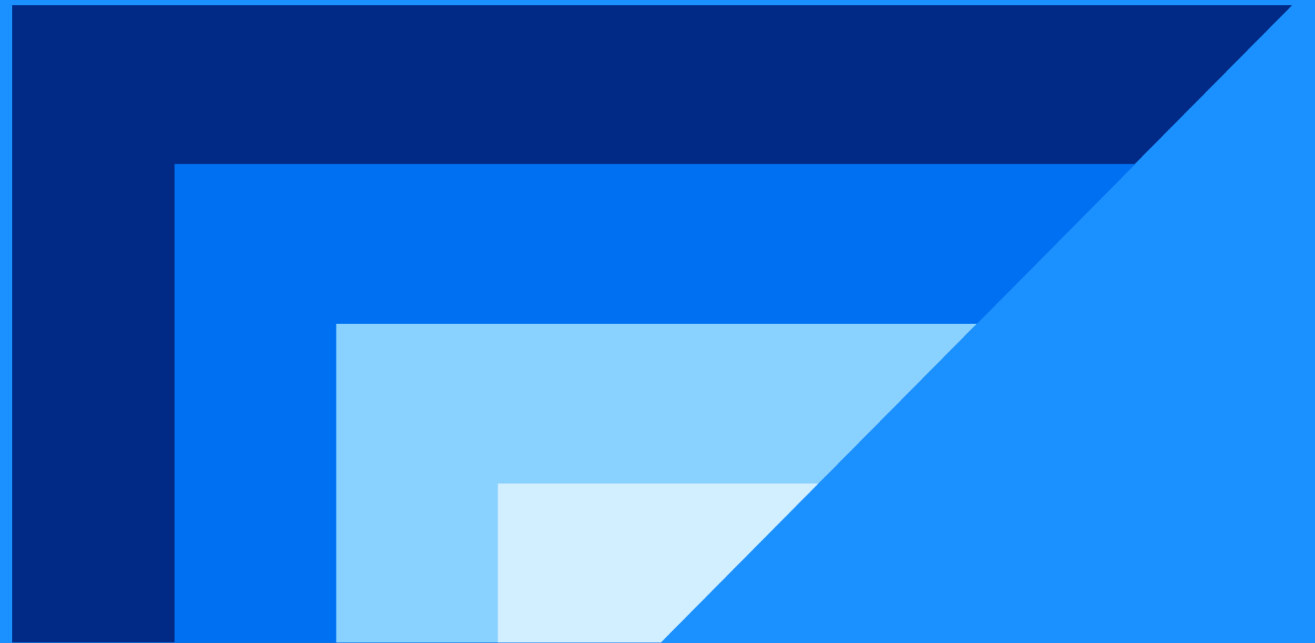
Digital workspaces

Joule skills and AI agents

SAP S/4HANA Cloud extensibility patterns



THE ABAP CLOUD DEVELOPMENT MODEL



ABAP Cloud



Is the ABAP **development model** to build cloud-ready business apps, services, and extensions

Supports transactional, analytical, and integration scenarios

Key user and developer extensibility options available

Is clean core compliant by definition

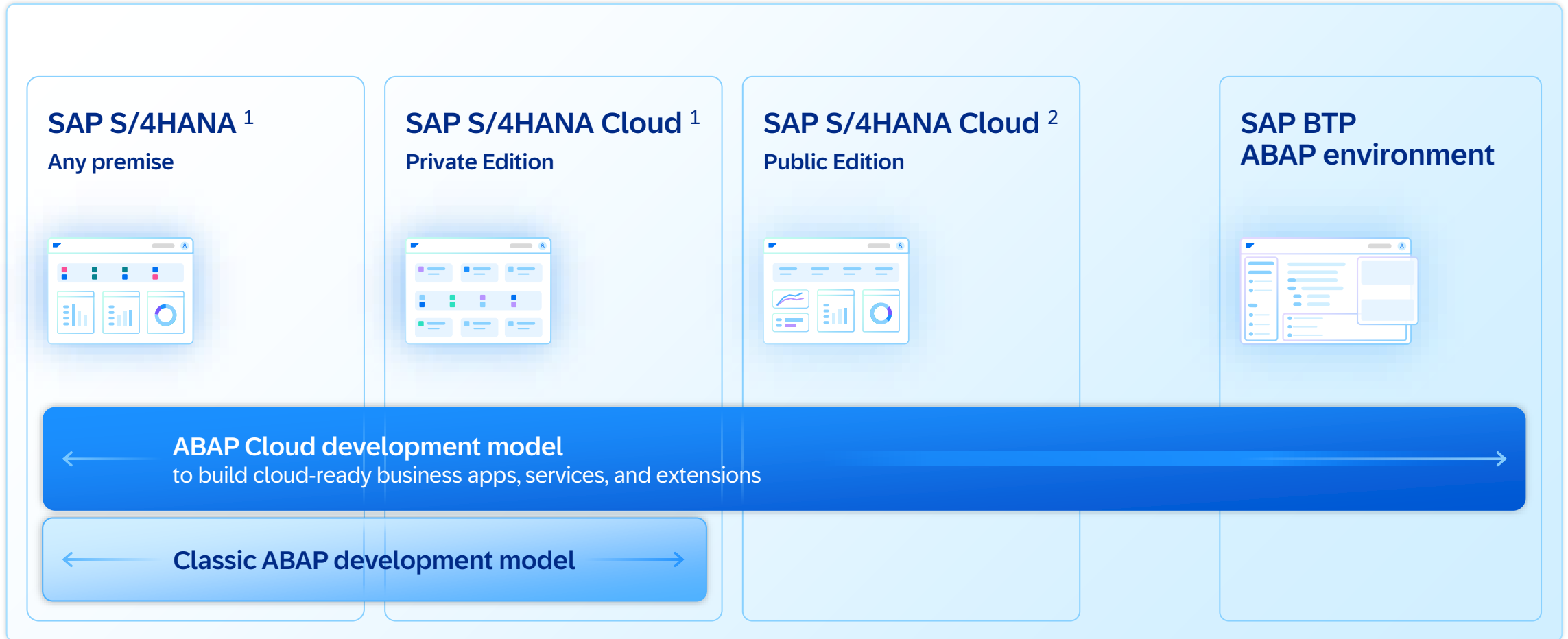
Is an integral part of SAP Build

Comes with SAP Business Technology Platform (SAP BTP) and SAP S/4HANA

Works with public or private cloud, and even on-premises

ABAP Cloud

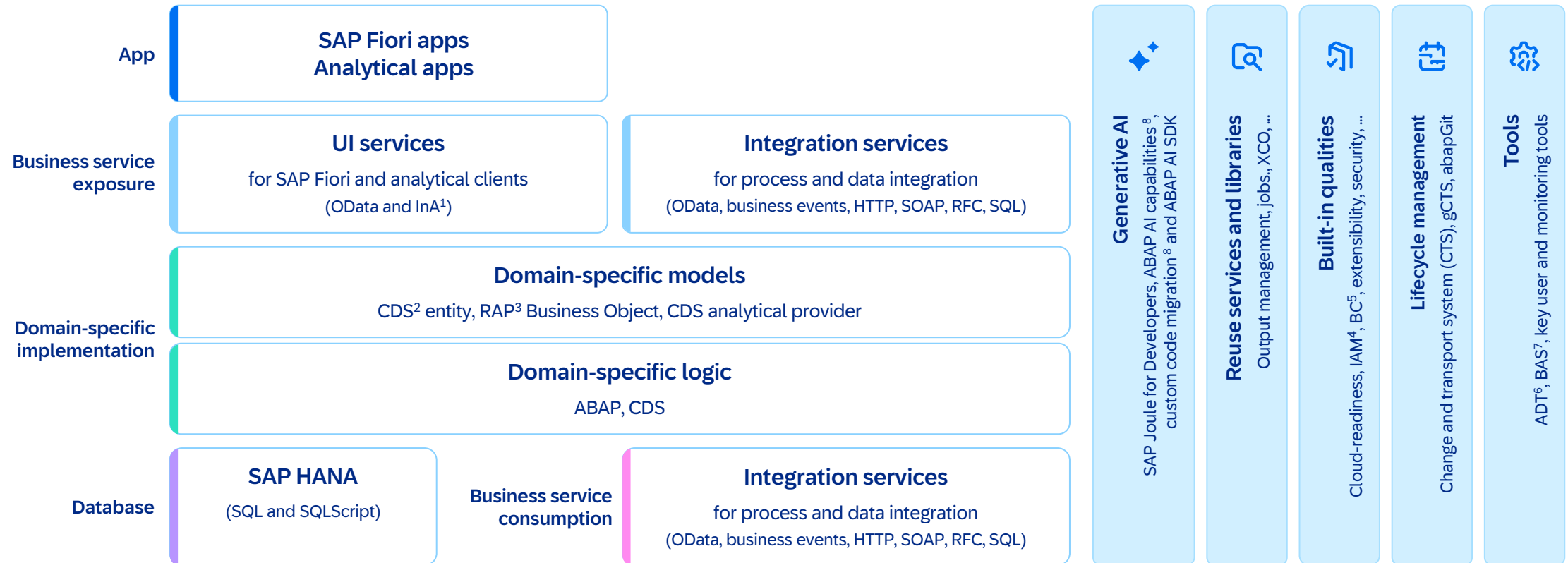
One development model for SAP BTP ABAP environment and all SAP S/4HANA editions



¹ SAP S/4HANA any premise or SAP S/4HANA Private Edition release ≥ 2022

² SAP S/4HANA Public Edition release ≥ 2208, 3-system landscape required

ABAP Cloud – High-level abstraction model



¹ SAP Information Access / ² Core Data Services / ³ ABAP RESTful application programming model / ⁴ Identity & Access Management / ⁵ Business Configuration / ⁶ ABAP development tools for Eclipse

⁷ Business Application Studio / ⁸ Available for SAP S/4HANA Cloud Private Edition as of 2021

ABAP PLATFORM 2025

PRODUCT HIGHLIGHTS





GENERATIVE AI IN ABAP

ABAP Platform 2025 feature scope

Generative AI

ACCELERATE: SAP JOULE FOR DEVELOPERS, ABAP AI CAPABILITIES

Joule copilot

Predictive code completion

ABAP code data services (CDS) code explanation

ABAP CDS unit test generation

ABAP Cloud application generation

Predict business logic in RESTful application programming model

OData service consumption

EMPOWER: ABAP AI SDK POWERED BY INTELLIGENT SCENARIO LIFECYCLE MANAGEMENT

Central, standardized access to the generative AI hub in SAP AI Core

Used by SAP for ABAP AI functionality

Integrate AI capabilities into custom ABAP applications and extensions

TRANSFORM: CUSTOM CODE MIGRATION AI ASSISTANT

Embedded in ABAP development tools for Eclipse (Joule copilot)

Docs chat for custom code migration to SAP S/4HANA

Explain legacy code (reports)

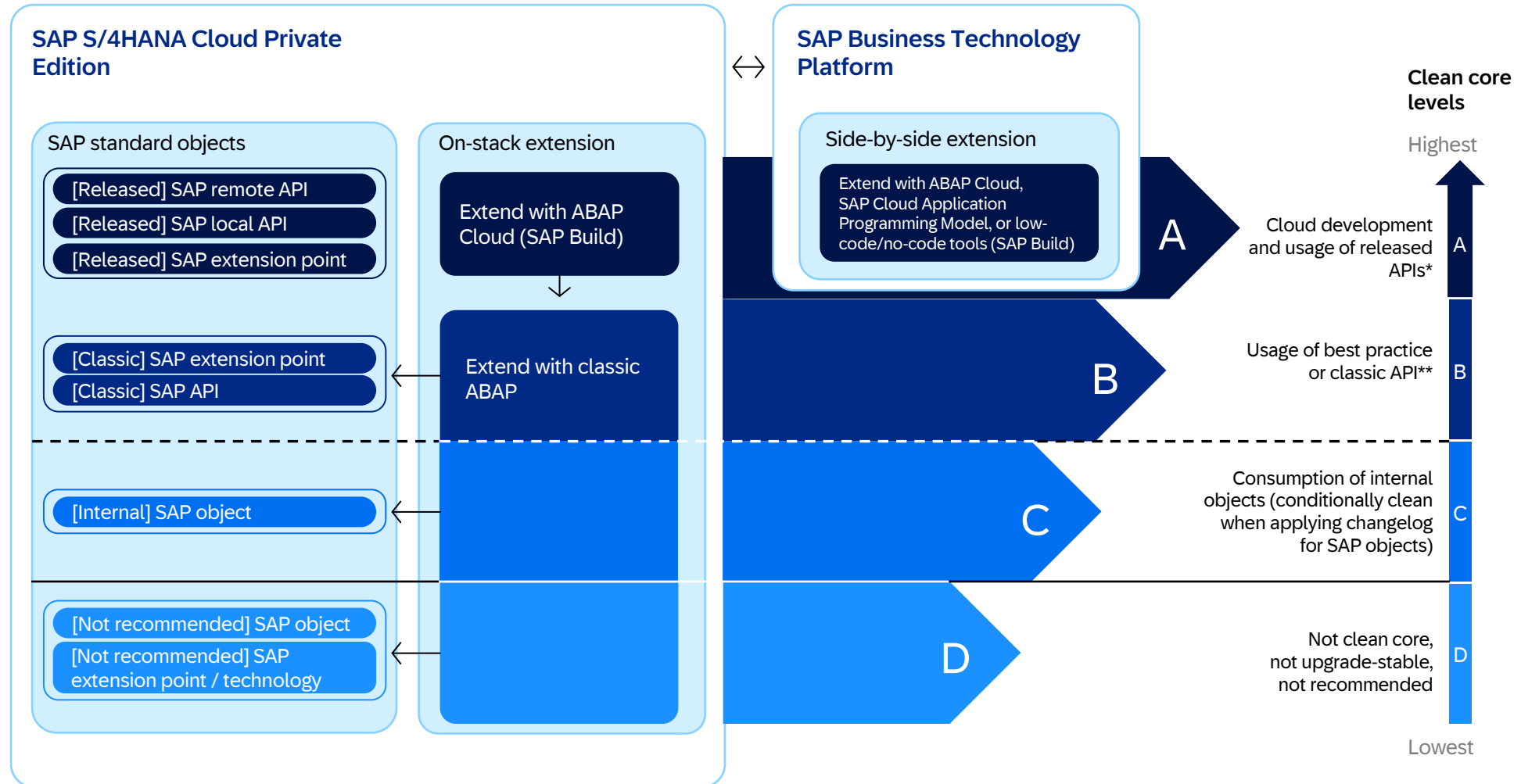
Explain ABAP test cockpit findings (SAP S/4HANA simplification issues)

Code proposal for code adaptation (with SAP S/4HANA Cloud Private Edition 2025 FPS01)

A light blue geometric shape, resembling a parallelogram with a diagonal cut, serves as a background for the title text.

CLEAN CORE DEVELOPMENT

Refined clean core extensibility model



[SAP Build](#) includes AI-enabled pro-code and low-code tools for on-stack and side-by-side application development and process automation.

* Available on SAP Business Accelerator Hub

** Available in cloudification repository viewer

Clean core level concept – What to do?



Read the [Clean Core Extensibility White Paper](#)



Read the revised [ABAP Extensibility Guide](#)

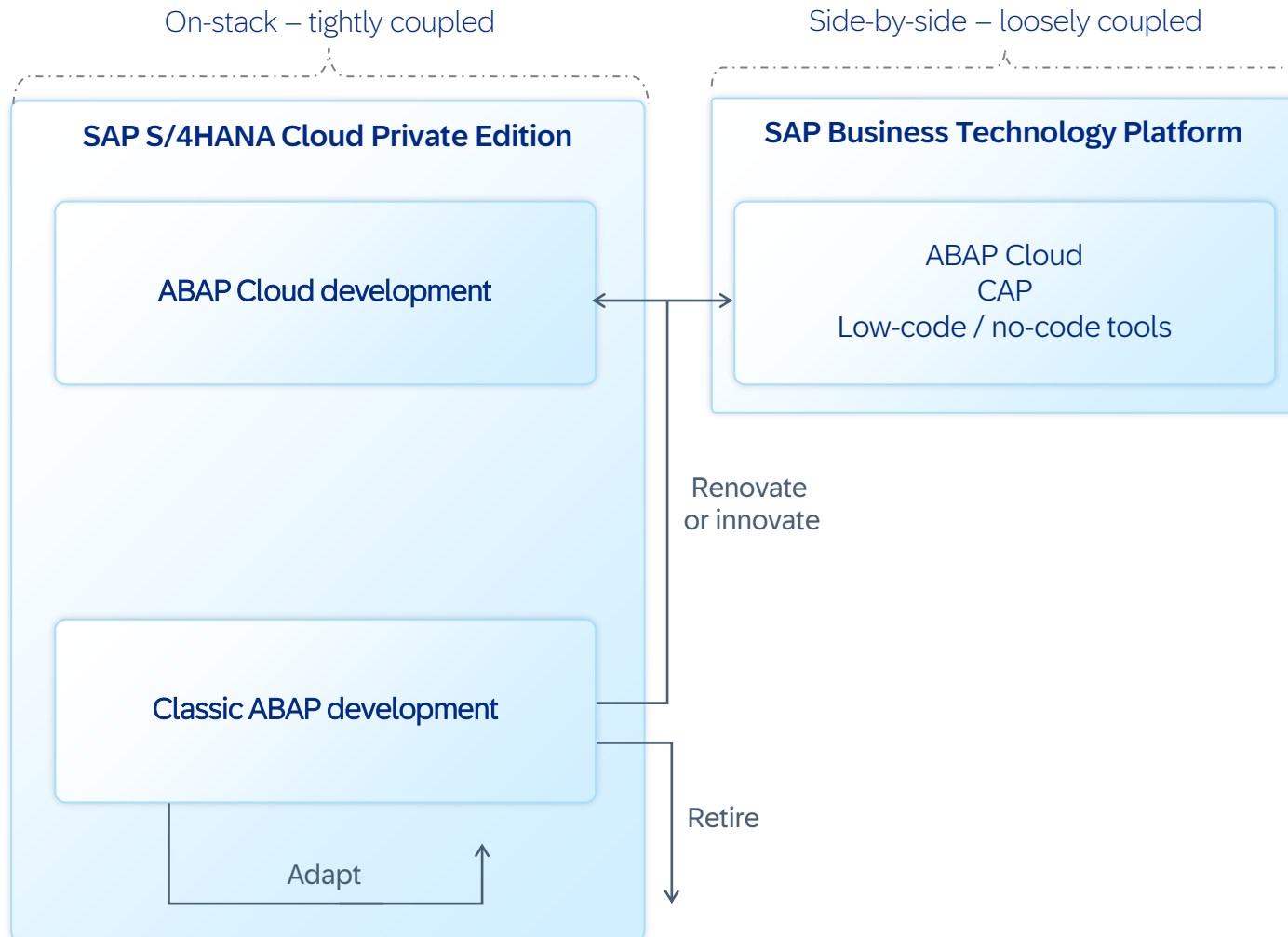


Read recommendations for the [ATC setup for clean core ABAP development](#)

- Adapt your clean core governance and your custom ABAP development guidelines
- If you started already with the 3-tier extensibility model: no problem, no disruption, but better guidance for classic ABAP
- Check the roadmap for updates (e.g. the changelog for SAP objects)
- Partners: Clean Core certifications are adapted to the new level concept. Contact the SAP Integration and Certification Center (ICC) for more information

Clean core extensibility for SAP S/4HANA Cloud Private Edition and on-premise

Stepwise cloud transformation



EXTENSIBILITY GUIDELINES



ABAP CLOUD DEVELOPMENT

Default choice for all new extensions, **clean core compliant** and cloud-ready, clear technical guidance

Mitigate missing released SAP APIs by implementing custom wrappers (**released custom APIs**)

CLASSIC ABAP DEVELOPMENT

For existing custom code, that can not be transformed to ABAP Cloud

For new extensions in application areas where no released SAP APIs or SAP Fiori UIs are available

Use **classic APIs** as foundation for developments

The list of recommended classic technologies and reuse services is provided with the SAP Note [3578329](#)

Governance of clean core principles: ABAP test cockpit (ATC), authorizations

/ Check out [ABAP Extensibility guide](#)

How to control the usage of released SAP APIs

ABAP Cloud development

By using ABAP language version (syntax-check)

ABAP Language Version: ABAP for Cloud Development

```
17 class zcl_business_partner implementation.  
18 method get_business_partner.  
19  
20 select single * from but000  
21 where type = '1' and partner = @bupa_id  
22 into @data(business_partner) .  
23  
24 endmethod.  
25 endclass.
```

ABAP Problems (5 items)

- The use of Data Element BU_PARTNER is not permitted.
- The use of Data Element BU_PARTNER is not permitted.
- The use of Table BUT000 is not permitted. Use CDS Entity I_BUSINESSPARTNER instead.
- The use of Table BUT000 is not permitted. Use CDS Entity I_BUSINESSPARTNER instead.
- The use of Table BUT000 is not permitted. Use CDS Entity I_BUSINESSPARTNER instead.

Classic ABAP development

By using ABAP test cockpit (ATC)

ABAP Language Version: Standard ABAP

```
17 class zcl_business_partner implementation.  
18 method get_business_partner.  
19  
20 select single * from but000  
21 where type = '1' and partner = @bupa_id  
22 into @data(business_partner) .  
23  
24 endmethod.  
25 endclass.
```

Description	Ref. Object
Findings: 2 Errors	
Errors (2 Errors)	
Usage of API that will not be released.	BUT000
Usage of not released ABAP Platform APIs.	BU_PARTNER

Details of Analysis

- Usage of API that will not be released. The use of Table BUT000 is not permitted. Use CDS Entity I_BUSINESSPARTNER instead.

Find released SAP APIs and classic SAP APIs

IN THE ABAP DEVELOPMENT TOOLS FOR ECLIPSE

- Using ABAP repository trees – e.g., **Released Objects** – and the **Search and Open ABAP Development Object** dialogs ([link](#))

ON THE SAP BUSINESS ACCELERATOR HUB

- [SAP S/4HANA Cloud Public Edition](#)
- [SAP S/4HANA Cloud Private Edition](#)
- [SAP S/4HANA](#)

USING ABAP TEST COCKPIT (ATC) WITH THE CLOUDIFICATION REPOSITORY ON GITHUB

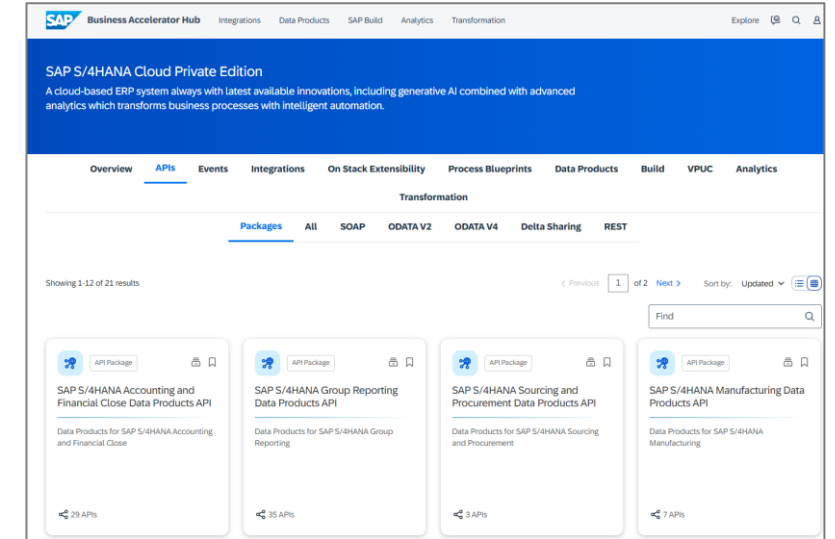
- [Cloudification Repository at SAP/abap-atc-cr-cv-s4hc \(github.com\)](#)

USING THE CLOUDIFICATION REPOSITORY VIEWER

- [Cloudification Repository Viewer at SAP/abap-atc-cr-cv-s4hc /github.com\)](#)

REQUEST THE RELEASE OF NEW SAP APIs IN SAP S/4HANA CLOUD

- SAP Customer Influence:
[Public Cloud Edition](#) | [Public Cloud Edition - APIs for Integration](#) | [What's New](#)



Application Component	Object Key	Object Type	State
BC-SRV-AP	APICL_BGRFC_CLEANUP_UTILS	CLAS	Classic API
BC-SRV-AP	APICL_ENABLER_DOC	CLAS	Classic API
BC-SRV-AP	APICL_ENABLER_PROXY	CLAS	Classic API
BC-SRV-AP	APICL_ENABLER_SYNC_LOG	CLAS	Classic API
BC-SRV-AP	APICL_ENABLER_XML	CLAS	Classic API
BC-SRV-AP	APICL_NRT_CONFIG_BASE	CLAS	Classic API
BC-SRV-AP	APICL_NRT_TRANSFER_PROCESS	CLAS	Classic API
BC-SRV-AP	APICL_NRT_TRANSFER_SIMPLE	CLAS	Classic API
BC-SRV-AP	APICL_PERS_QUEUE	CLAS	Classic API
BC-SRV-AP	APICX_AP_ENGINE_BASE	CLAS	Classic API
BC-SRV-AP	APICX_AP_ENGINE_NOT_FOUND	CLAS	Classic API
BC-SRV-AP	APICX_ENABLER_BASE	CLAS	Classic API

How to mitigate missing publicly released SAP APIs

CUSTOM WRAPPER DEVELOPMENT



Classic APIs are the foundation for custom wrappers

[Classic APIs list](#) available on GitHub

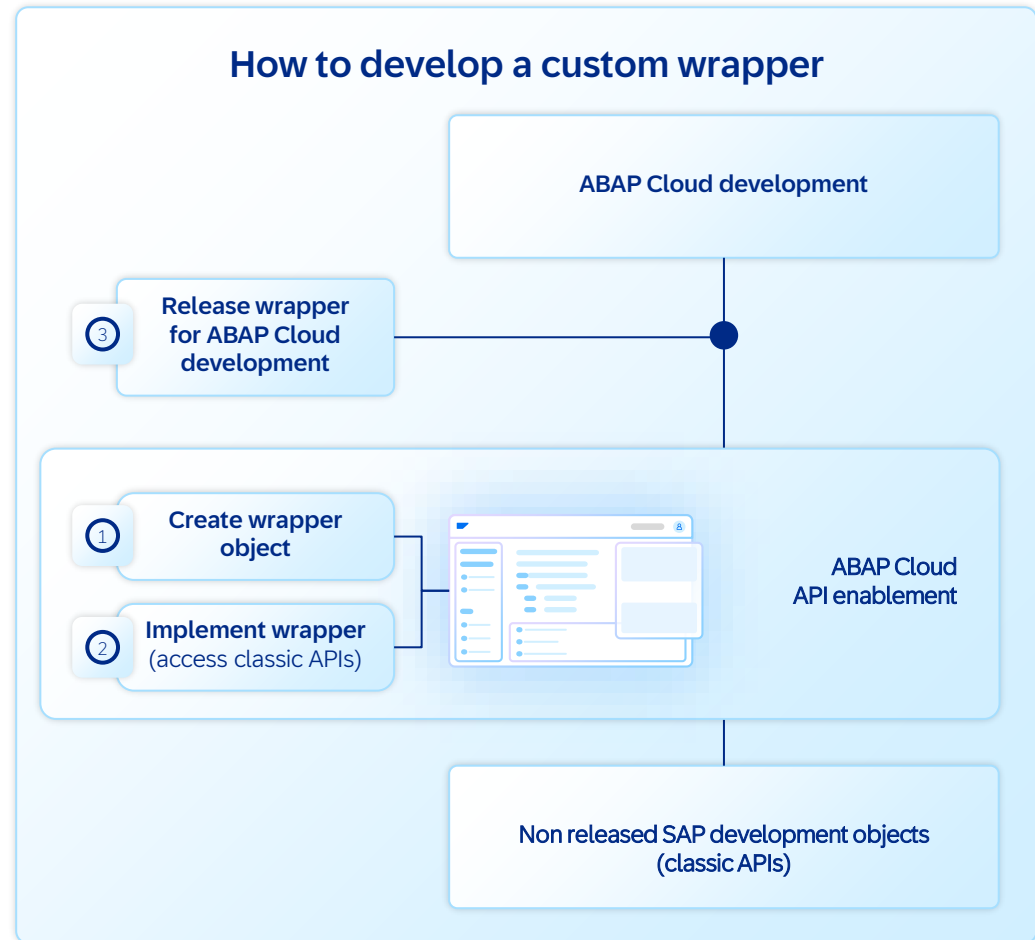
[Cloudification repository viewer \(sap.github.io\)](#) available

Usage of classic APIs can be checked by ABAP test cockpit

Custom wrappers for objects using classic APIs can be implemented

[Custom wrapper generator](#) available for function modules (BAPIs)

How to develop a custom wrapper



Use ABAP test cockpit (ATC) to govern clean core

Set up ATC

[Remote ABAP test cockpit](#): Connect DEV systems to the central ATC system

Configure ATC to run during development tasks and at transport release

Block priority 1 and priority 2 ATC findings for transport

Set up ATC check variant

If using ATC on SAP BTP (and with upcoming SAP S/4HANA 2025 FPS01 release or beyond)

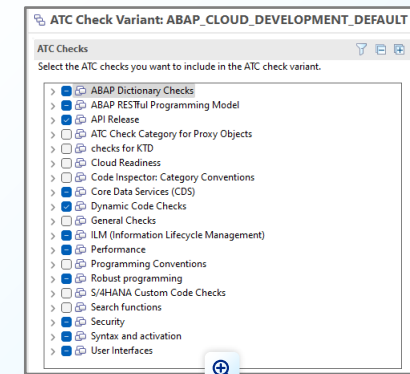
[Use the variant ABAP_CLEAN_CORE_DEVELOPMENT](#)

If using ATC in SAP S/4HANA 2025 FPS00 system:

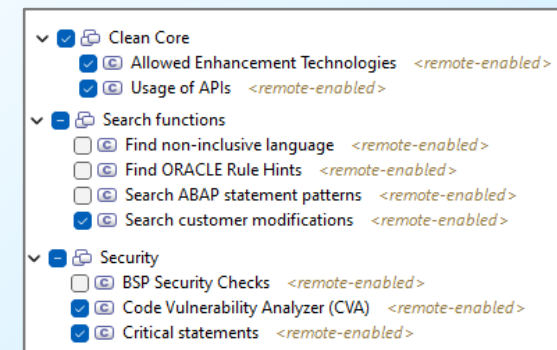
[Copy the variant ABAP_CLOUD_DEVELOPMENT_DEFAULT](#) and add the following checks

- Allowed SAP enhancement technologies *
- Usage of APIs *
- Critical statements
- Search for customer modifications
- [Code Vulnerability Analyzer \(CVA\)](#) (if CVA is licensed accordingly)

On-premise: Copy ATC variant ABAP_CLOUD_DEVELOPMENT_DEFAULT



Add checks



Suppress findings in legacy objects that cannot be adopted with ATC baseline

Request ATC exemptions if SAP internal objects are still needed to control their usage

New ATC check based on usage guidelines for SAP APIs

New ABAP Test Cockpit check

☒  Usage of APIs <remote-enabled>



Check Behavior

No ATC finding

- Usage of released API

Priority 3 (Info)

- Usage of classic API

Priority 2 (Warning)

- Usage of internal APIs (not classified API)
- SQL read access to SAP database table
- SUBMIT statement on programs

Priority 1 (Error)

- Usage of “no API” (with successor if available)
- SQL write access to SAP database table
- PERFORM statements on external subroutines

Check a simple ABAP report with ALV using

ABAP Cloud readiness checks:

Findings: 14 Errors	
ABAP Language Version (Syntax) (5 Errors)	
Syntax error in restricted language scope (Open SQL)	SELECT
Syntax error in restricted language scope (Open SQL)	UPDATE
Syntax error in restricted language scope (dynpro)	PARAMETERS
Syntax error in restricted language scope (report)	REPORT
Syntax error in restricted language scope (report)	START-OF-SELECTION
Allowed Object Types in Cloud Development (1 Errors)	
Objects of type PROG are not allowed in ABAP Cloud Development	
Usage of Released APIs (8 Errors)	
Usage of API that will not be released.	BUT000
Usage of API that will not be released.	BUT000
Usage of API that will not be released.	BUT000
Usage of API that will not be released.	BUT000
Usage of not released ABAP Platform APIs.	BU_PARTNER
Usage of not released ABAP Platform APIs.	CL_SALV_TABLE
Usage of not released ABAP Platform APIs.	CL_SALV_TABLE
Usage of not released ABAP Platform APIs.	CX_SALV_MSG

Usage of APIs check:

Findings: 1 Errors, 1 Warnings, 3 Infos	
Errors (1 Errors)	
Updating DDIC database tables or DDIC table views is not allowed (successor available)	BUT000
Warnings (1 Warnings)	
Reading from DDIC database tables or DDIC table views is not recommended (successor available)	BUT000
Infos (3 Infos)	
Usage of classic API	CL_SALV_TABLE
Usage of classic API	CL_SALV_TABLE
Usage of classic API	CX_SALV_MSG

/ Learn more : [ATC recommendations for governance of clean core development](#)

How ABAP test cockpit supports the clean extensibility level concept

CLEAN EXTENSIBILITY LEVELS

	Usage of SAP objects in custom development	Clean core checks in ABAP test cockpit *
A	Released SAP APIs (local and remote) and extension points	No messages
B	Classic SAP APIs and extension points	Priority 3: Info message
C	Internal SAP objects	Priority 2: Warning message
D	Not recommended SAP objects and extension technologies	Priority 1: Error message

* The checks **Usage of APIs**, **Allowed Enhancement Technologies**, and **Critical Statements** are considered for the clean extensibility level determination of customer objects.

/ Learn more : [Clean Extensibility for SAP S/4HANA Cloud](#) white paper and the [ABAP extensibility guide](#)

ABAP Platform 2025 feature scope

Custom code management

ATC CHECKS CLEAN CORE

New ATC checks of category Clean Core: Usage of APIs and Allowed Enhancement Technologies

CUSTOM CODE MIGRATION APP

Include ADT-Links in export of ATC result for the SAP S/4HANA Readiness Check for navigation to the source code

Provide options to start analysis automatically at project creation or manually after project creation

Scope calculation must be now initiated manually after project creation

FOR USE WITH REMOTE ATC ON SAP BTP

Update accessible RFC function modules for the latest release of SAP BTP ABAP environment
(import ZIP file attached to SAP Note [2861842](#))

EXTEND ATC

CDS views to access ATC results created by Analyze Custom Code app/Custom Code Migration app
(SYCM_APS_PROJECT, SYCM_APS_ATC_FINDINGS)

see also [How to extend ABAP Test Cockpit on SAP BTP for custom use cases](#)

ABAP test cockpit on SAP BTP

ABAP test cockpit on SAP BTP is recommended for static quality assurance of private cloud and on-premise landscapes

ABAP test cockpit on SAP BTP is available in the SAP BTP ABAP environment



Always up-to-date check system operated and maintained by SAP

Custom code analysis including security checks
(No additional license fee required for SAP Code Vulnerability Analyzer)

Usage data consideration (SCMON, SUSG)

Custom ABAP test cockpit check variants and custom checks

Developer scenario

Robust workflow for handling of ABAP test cockpit exemptions

Baseline to exclude ABAP test cockpit findings in old legacy code

Scheduling of ABAP test cockpit runs

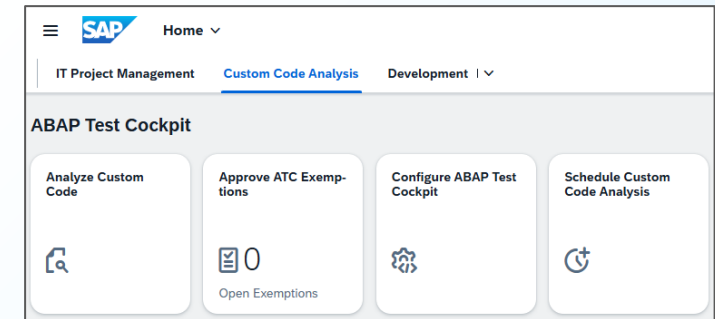
Transport management and integration of change request management (ChaRM)

Customer Success Stories

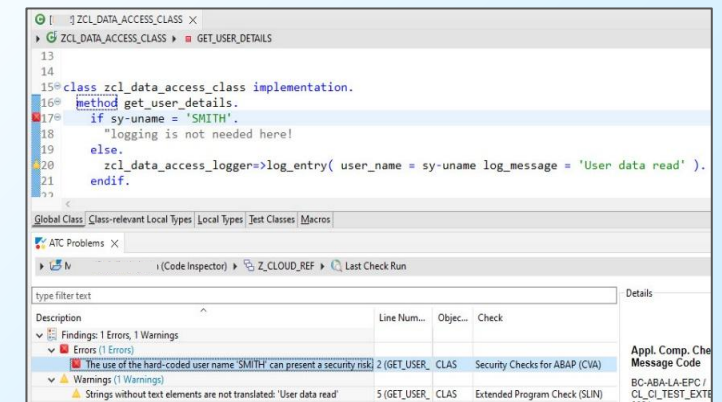
Geberit: [Improve code quality with ATC on SAP BTP](#)

REWE Digital: [Build clean core apps with SAP BTP ABAP environment and ATC on SAP BTP](#)

ABAP test cockpit on SAP BTP



ABAP test cockpit developer scenario in ABAP development tools for Eclipse



/ Check out [ABAP test cockpit on SAP BTP - Overview](#)

A light blue geometric shape, resembling a parallelogram with a diagonal cut, serves as a background for the title text.

ABAP DEVELOPMENT TOOLS FOR ECLIPSE

ABAP Platform 2025 feature scope

ABAP development tools for Eclipse

DEVELOPMENT & PRODUCTIVITY

- Refactoring ABAP classes and interfaces
- New where-used dialog with prefiltering of results
- New property **abaplv** to filter searches with ABAP language version
- Clear console when running an ABAP application
- Software component relation editor
- Manage business (electronic) document consistency
- Manage intelligent scenarios and intelligent scenario models for AI solutions
- Maintain Adobe Form templates

UNIT TESTING

- Use quick assists to insert ABAP unit test class templates
- Launch ABAP unit tests via code minings
- Display failed assertions and status of test methods directly in the code
- Rerun unit tests with coverage
- Test code highlighting for method calls

AI

- Integration of SAP Joule for Developers, ABAP AI capabilities

EXTEND IDE WITH CUSTOM ACTIONS

- Develop custom actions with the IDE action framework and enhance them with server-driven UI

RAP & CDS

- Create list report-based SAP Fiori applications directly from the service binding using the Quick SAP Fiori application generator
- Launch SAP Fiori project wizard in VS Code/BAS directly from service binding
- Create CDS table entities
- Create CDS aspects to store field definitions and calculations for reuse
- Generate extension fields to existing RAP BO using wizard
- CDS external entities are now supported in Data Preview
- Filter the log when migrating CDS DDIC-based views to CDS view entities
- Triggering where-used list for usages of CDS annotations

ANALYTICS

- Generate review booklets apps and multi-dimensional analysis SAP Fiori apps

DEBUGGING & MEMORY ANALYSIS

- ABAP debugger: Displaying component groups in the variables view
- ABAP debugger: Variables with hidden values (for sensitive data)
- ABAP debugger: Table compare tool
- Memory: Displaying memory consumption of a debugged ABAP program
- Memory: Create ABAP memory snapshot

TRANSPORT MANAGEMENT

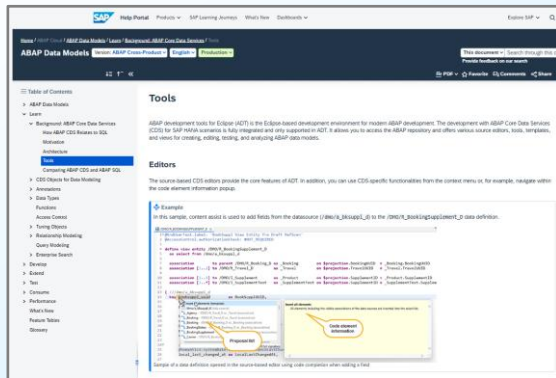
- Enhancements to adding an object to a transport
- Searching transport requests
- Grouping requests in transport organizer

ABAP development tools for Eclipse (1)

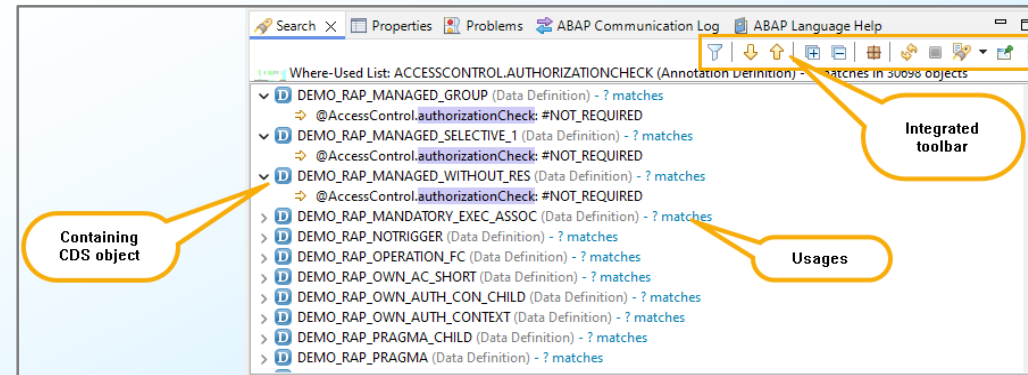
RAP & CDS enhancements

Explore available ABAP CDS Tools

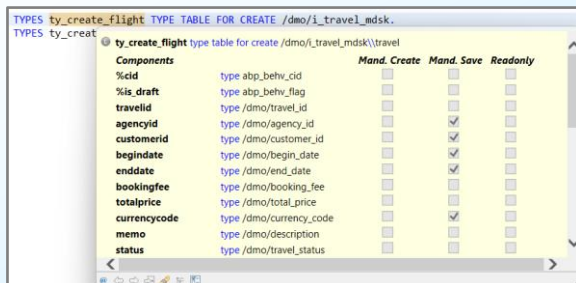
<https://help.sap.com/docs/abap-cloud/abap-data-models/tools>



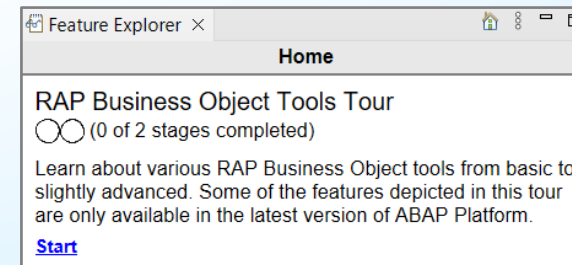
Trigger where-used list for usages of CDS annotations



Display field characteristics for RAP types in the element info popup



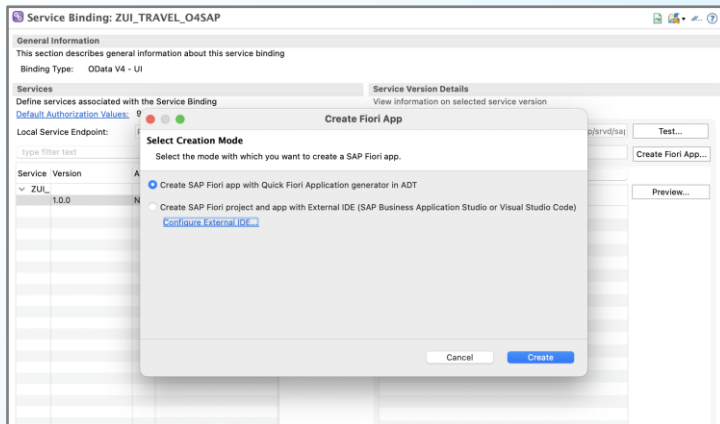
Explore RAP BO development features with the new Feature Explorer tour



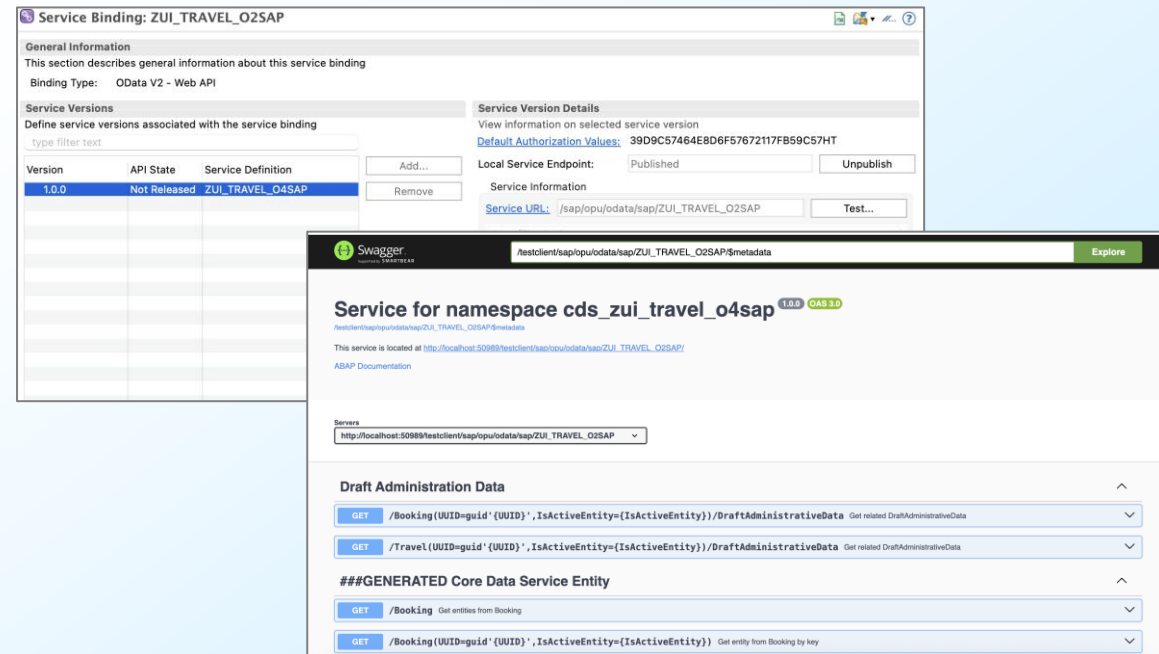
ABAP development tools for Eclipse (2)

Working with business services

List Report based SAP Fiori application can be created from service binding using the Quick SAP Fiori application generator



Web API-based services can be tested from service binding using swagger UI



ABAP development tools for Eclipse (4)

ABAP debugger enhancements and further troubleshooting improvements

ABAP debugger preview with ABAP memory consumption

Current debugger position

ABAP Memory view

Display of the respective memory consumption in accordance to the current debugger position

Choose debugger step size

Garbage Collector
Database Commit
Database Rollback
Create Memory Snapshot
Update Debugging
• Step Size: 'Line'
Step Size: 'Expression'

Display warnings sent by SAP HANA in the AMDP debugger

AMDP Debugger Warning

Debugging of scalar UDFs is not fully supported yet!...

Show Full Text

ABAP debugger preview with ABAP memory consumption

STREET

FURTHER IMPROVEMENTS

Collaborative error analysis:

Export and import ABAP cross trace result to/from JSON file

Better handling of large ABAP traces: Minimize and restructure the number of levels in the aggregated call tree

Exporting short dump as ZIP file in the ABAP runtime error viewer

ABAP Debugger:

Format and display variable types of JSON in the variables view

ABAP LANGUAGE

ABAP Platform 2025 feature scope

ABAP language (1)

ENHANCEMENTS FOR INTERNAL TABLES

The addition `WHERE` can now be used with the `READ TABLE` statement to read the first line of an internal table that fulfills the `WHERE` condition

Supporting arbitrary logical expressions as well as equality conditions (to be specified)

The predicate `IS [NOT] INITIAL` is now possible in the `WHERE` condition of the basic form of the `FILTER` operator

Harmonization of table expressions and `READ TABLE` statement

NEW ABAP SYSTEM CLASSES FOR PROBABILITY DISTRIBUTION [\[Link\]](#)

Enable the generation of random number and the calculation of quantile, probabilities, and other aspects:

- `CL_ABAP_PROB_DISTRIBUTION` to define probability distributions in ABAP for probabilities of a floating-point data type
- `CL_ABAP_PROB_DISTRIBUTION_DF34` to define probability distributions in ABAP for probabilities of the built-in data type `decfloat34`

Various discrete and continuous distributions supported: `normal`, `log-normal`, `gamma`, `exponential`, and uniform `int/int8` distributions.

ENHANCEMENTS FOR TRANSFORMATIONS

Regular expression functions can now be used in XSLT – available are `matches()`, `replace()`, and `tokenize()`

New domain `XSDGEO` for serializing and deserializing spatial data between the Extended Well-Known Binary (EWKB) representation and the GeoJSON format

ST statement `tt:value` allows new formats to be specified in parentheses after format in the option attribute: `geometry`, `geography`, and `geo=SRID` to define a spatial reference system for the conversion of the EWKB representation to GeoJSON and vice versa

ABAP Platform 2025 feature scope

ABAP language (2)

ENHANCEMENTS FOR EXPRESSIONS AND FUNCTIONS

Direct assignment of string expressions in **VALUE** operator without requiring the use of **CONV** to structure components

New built-in factorial and binomial functions to compute the factorial of positive integer values and the binomial coefficient, respectively

New arithmetic functions for statistics and probability theory in ABAP - for calculations with floating point numbers and decimal floating points numbers:

- **ERF**, **ERFC**, and **ERF_INV** to implement the error function, the complementary error function and its inverses, respectively.
- **GAMMA** and **LOG_GAMMA** to implement the gamma and log gamma functions, respectively
- **GAMMA_LOWER** to implement the lower incomplete gamma function and its inverse

NEW ADDITION FORWARDING PRIVILEGED IN ABAP EML

The privileged context can be propagated in subsequent ABAP EML requests using the **FORWARDING PRIVILEGED** addition in **READ**, **MODIFY**, and **GET PERMISSIONS** statements.

CHANGE FOR SAP GUI USER DIALOGS

Behavior of character ! in the initial position of input fields on dynpro screen can be controlled by the new profile parameter **dynp/disable_input_reset**: **TRUE** (disabling standard behavior) and **false** (preserve standard behavior)

/ Check out the [News for ABAP Release 8.16](#)

/ Check out [ABAP Language at Devtoberfest 2025](#)

A light blue geometric shape, resembling a parallelogram or a rectangle with a diagonal cut, serves as a background for the title text.

DATABASE-CENTRIC OPTIMIZATIONS

ABAP CDS DATA MODELS

CONSUMPTION

NATIVE CONSUMPTION

ABAP SQL

CDS

AMDP

EML

SERVICE CONSUMPTION

SAP Fiori UI

Analytical client

Web APIs

SQL

Business events

BUSINESS SERVICE EXPOSURE & DATA ACCESS

SERVICE BINDINGS

SERVICE DEFINITION

SERVICE PROJECTIONS AND INTERFACES

CDS projection views

DATA MODELING

TUNING OBJECTS

ABAP tuning objects

SAP HANA tuning objects

CDS DATA MODELS

CDS types

Domain specific data models

CDS modeling entities

CDS table functions for SAP HANA native
objects (procedures, functions)

CDS access
controls

EXTENSIBILITY

ABAP Platform 2025 feature scope

ABAP CDS

CDS table entities

CDS updateable view entities

CDS hierarchy generation wizard

Creation of local ABAP CDS models based on imported external models

CDS explain powered by AI

Improved CDS buffering

ABAP Platform 2025 feature scope

AMDP

CLIENT SAFETY OF AMDP METHODS

AMDP methods can be made client-safe to support the concept of client isolation

Client safety is mandatory for the language version for ABAP Cloud (ABAP for Cloud Development)

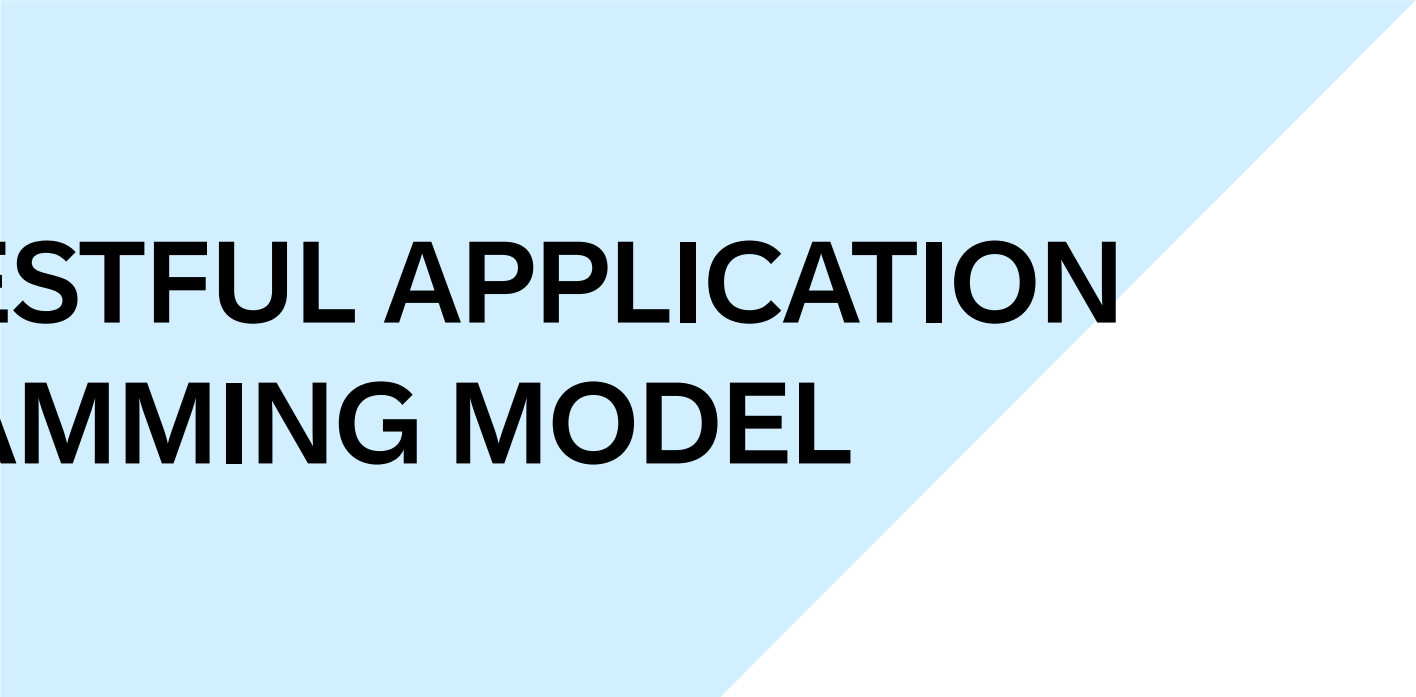
NEW AMDP OPTION CDS SESSION CLIENT DEPENDENT

The new addition `CDS SESSION CLIENT DEPENDENT` can be used after `AMDP OPTIONS` to set the HANA session variable `CDS_CLIENT` to its default value and to declare the AMDP method explicitly as client-dependent and to make it client-safe

The addition can be also used as an option after `FOR DDL`

NEW ADDITION FOR SQL SERVICE

New addition `FOR SQL SERVICE` to the statement `CLASS-METHODS` to specify that an AMDP procedure implementation is intended to be exposed for external SQL access using an SQL service

A light blue geometric shape, resembling a parallelogram or a trapezoid, is positioned in the center of the slide. It has a diagonal line running from the top-left corner to the bottom-right corner, creating two triangular sections within the shape.

ABAP RESTFUL APPLICATION PROGRAMMING MODEL

ABAP Platform 2025 feature scope

ABAP RESTful Application Programming Model (1)

HIGHER EFFICIENCY WITH NEW GENERATORS

New generators provided to generate RAP BOs with one or more entities and OData UI services from scratch, with and without AI assistance, and to generate custom UI service based on RAP BO interface

ENHANCED CDS SUPPORT FOR MANAGED BOs

CDS table entities as active persistence layer for managed RAP BOs without draft now supported. [\[Learn more\]](#)

ENHANCEMENTS FOR UI SERVICES FOR SAP FIORI ELEMENTS-BASED APPs

Collaborative draft handling – to allow multiple users to work simultaneously and collaboratively on the same draft instance of an entity without locking each other out

Event-driven RAP side effects – to allow asynchronous changes in SAP Fiori UIs. Whenever such an RAP business event is raised, the associated RAP side effect is triggered and the defined targets are reloaded in the UI

Read-only and editable tree views – to support for displaying hierarchical data in SAP Fiori UIs, with drag and drop, cut and paste, reorder, and cascading delete options

Support for SAP Fiori UIs with multiple RAP BOs – to support cross-BO scenarios where two or more RAP BOs interact with each other. Using cross-BO transactional enablement where the entities of the participating RAP BOs are linked through cross-BO associations

ABAP Platform 2025 feature scope

ABAP RESTful Application Programming Model (2)

ENHANCEMENTS FOR RAP BEHAVIOR DEFINITION (BDEF)

Definition of **RAP business events** for RAP BO child entities now possible

Specifying **RAP BO child entities as authorization master** now also supported

BDEF friends can be used for specifying one or more BDEF extensions for a base BDEF or for modularization purposes. They have unrestricted access to all components of the base BDEF.

ENHANCEMENTS FOR ABAP BEHAVIOR POOL

RAP default values function supported – as optional RAP BO operations used to default field values or input parameters for RAP BO operations on the UI. A RAP default values function must be implemented in a predefined handler method of the relevant ABAP behavior pool.

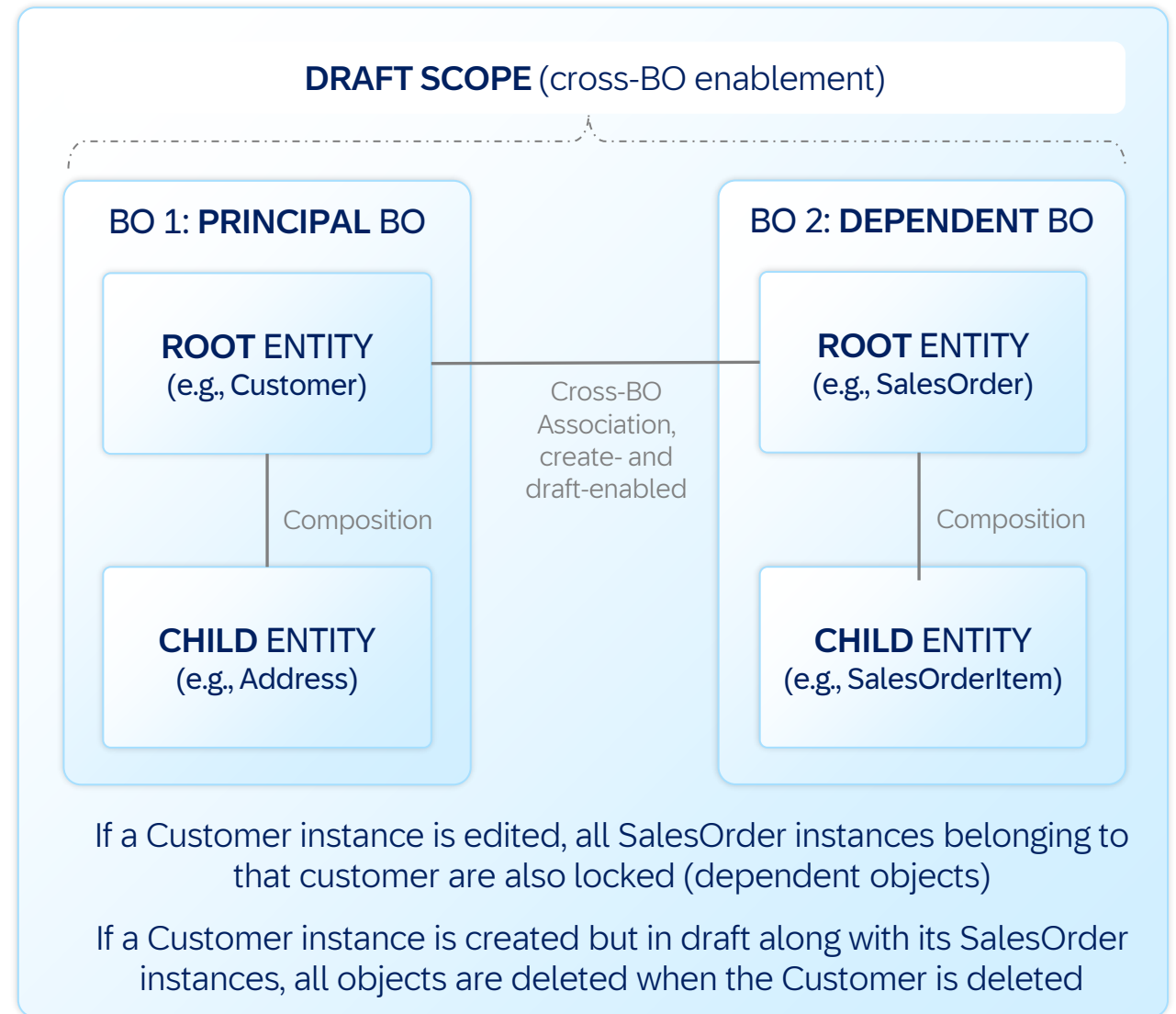
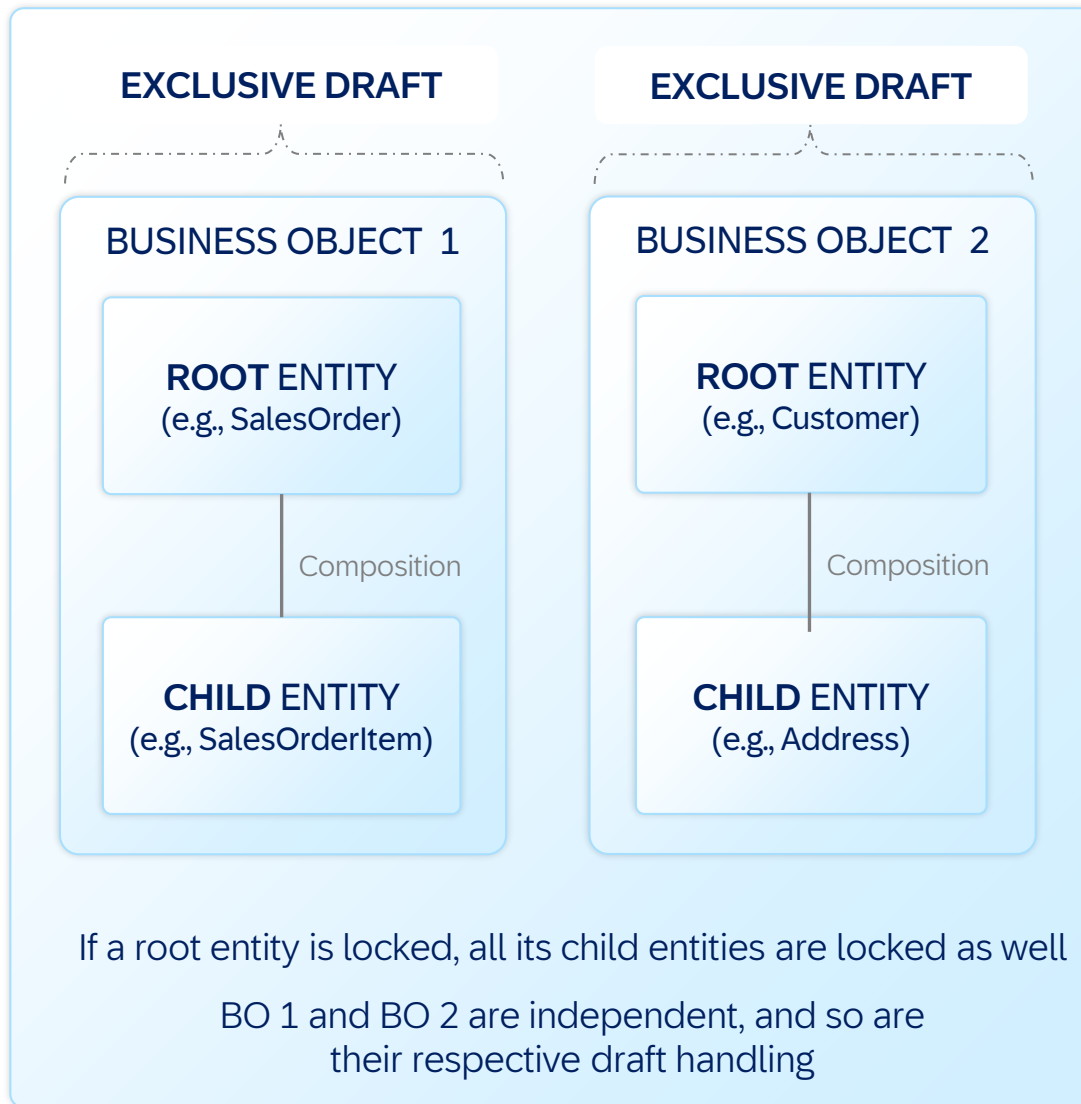
Auxiliary classes for ABAP behavior pools now possible – to enable extended access rights to a referenced RAP BO, with the primary purpose of providing reusable methods and allowing developers to implement RAP behavior concurrently. [\[Learn more\]](#)

C1-release of RAP saver class c1_abap_behavior_saver_failed for ABAP Cloud development – to allow REPORTED and FAILED to be filled in the late save phase when using BAPIs. [\[Learn more\]](#)

ABAP CONTRACT CHECK VIOLATIONS VISIBLE IN UNIT TESTS

Violations from ABAP contract checks, i.e. RAP BO contract checks and transactional contract checks, are made visible in unit tests, that is, the unit tests will fail if a contract check violation occurs.

Standard draft vs. draft scope



SAP Fiori elements tree table

USE CDS HIERARCHIES TO MODEL HIERARCHICAL STRUCTURES AND EXPOSE THEM AS TREEVIEWS TO THE SAP FIORI UI

In the context of RAP, a treeview visualizes data modeled by a CDS hierarchy using an SAP Fiori Elements tree table, for example

Examples for hierarchically business data structured in parent-child relationships

- Company or department structures with employees and their managers

- Material and product classifications, for example, for customs

- Manufacturing products with a hierarchical structure (such as the component parts of a car)

Employees (39,593)						
Employee		First Name	Last Name		Salary	Manager
✓ 1		Friedrich	Meier	128,147.24	EUR	
2		Johannes	Rahn	59,133.76	EUR	1
✓ 3		Lee	Neubasler	90,975.40	EUR	1
> 4		Holm	Illner	79,575.07	EUR	3
> 168		Andreas	Martin	84,177.44	EUR	3
> 725		Felix	Benz	72,435.73	EUR	3
> 787		Georg	Benz	61,311.15	EUR	3
✓ 798		Anna	Babilon	76,002.62	EUR	3
799		Guenther	Neubasler	52,552.62	EUR	798
> 800		Johann	Wohl	69,110.67	EUR	798
> 815		Martin	Koslowski	66,392.76	EUR	798

A light blue geometric shape, resembling a parallelogram with a diagonal cut, serves as a background for the title text.

EMBEDDED ANALYTICS

ABAP Platform 2025 feature scope

Embedded analytics

ABAP CLOUD GENERATOR WIZARDS

Create multidimensional analysis apps
(based on a cube- or dimension view)

Create Review Booklet apps
(based on a service binding of type InA)

Create native CDS hierarchy definitions
(based on a dimension view)

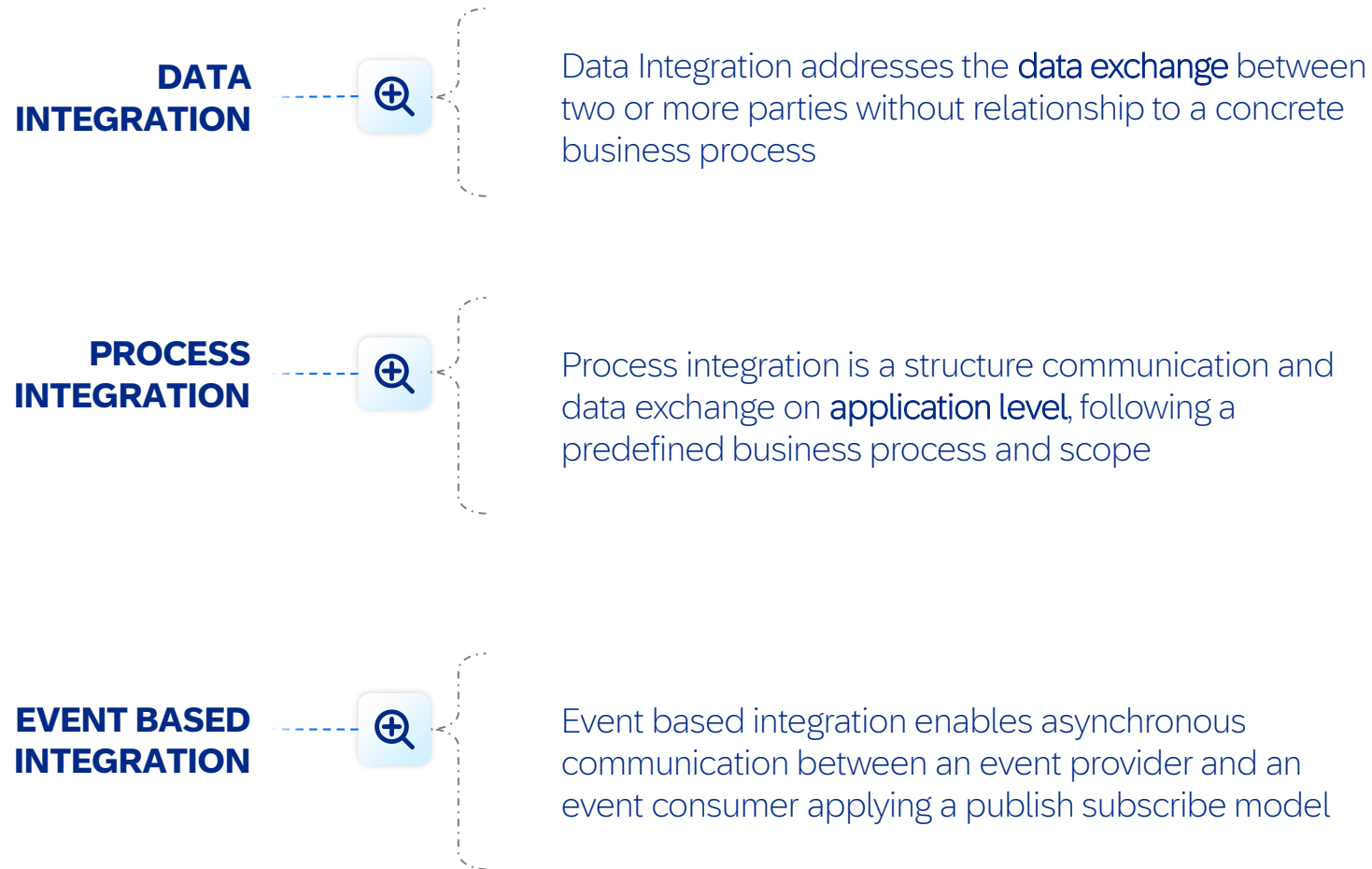
CDS REUSABLE MEASURES AND FILTERS VIA CDS ASPECTS

ANALYTICS PERFORMANCE OPTIMIZATIONS

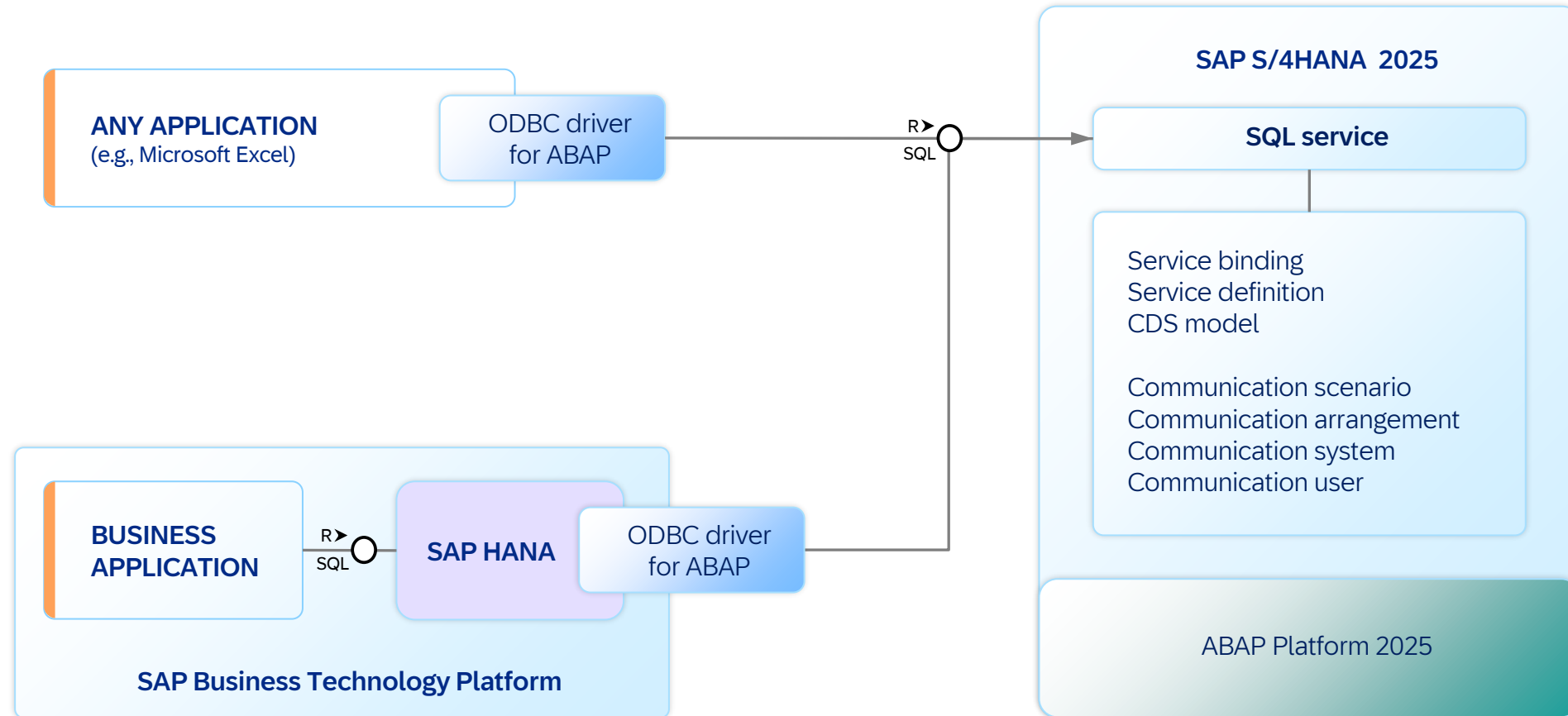
A light blue geometric shape, resembling a stylized arrow or a parallelogram with a diagonal cut, pointing towards the top right. It serves as a background for the main title.

INTEGRATION SERVICES

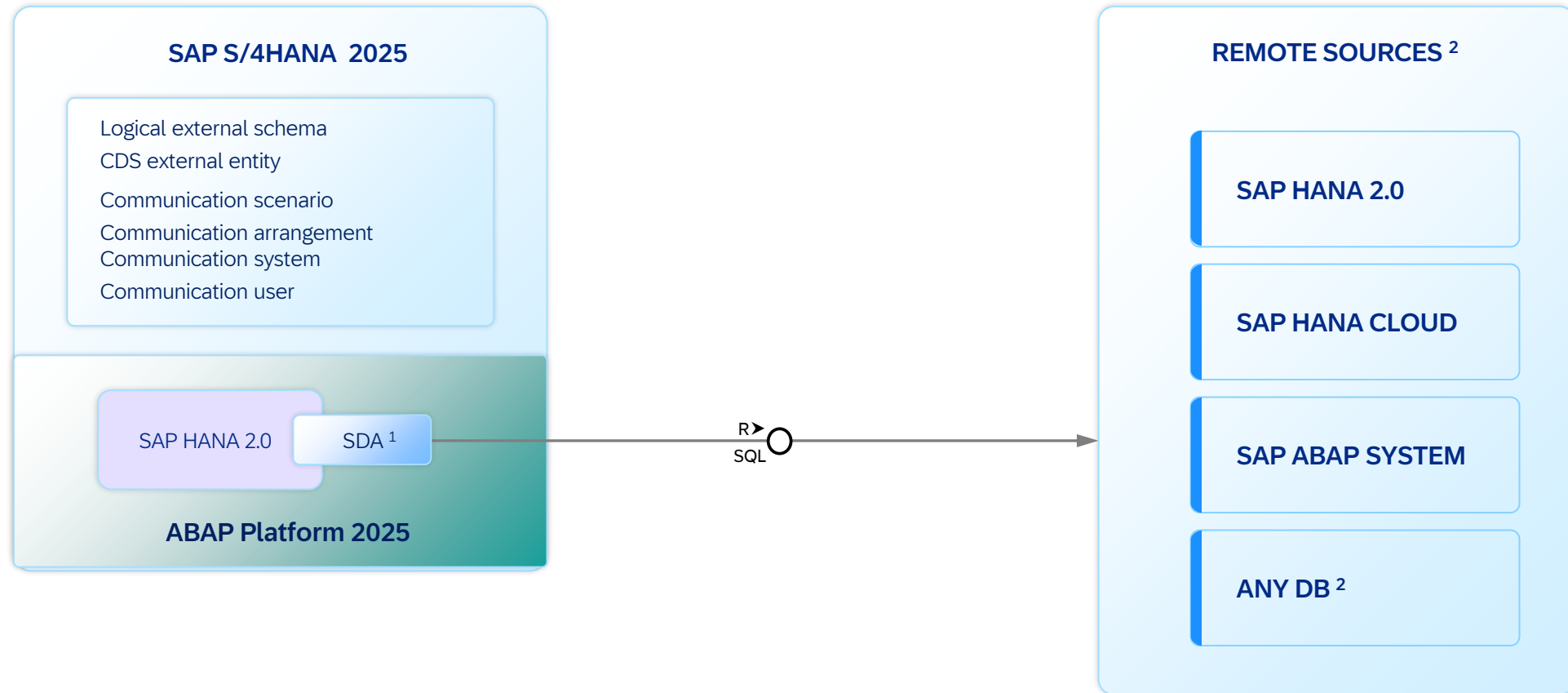
Integration scenarios



Data exposure based on ABAP SQL service



Data consumption based on CDS external entities



¹ Smart Data Access

² / [SAP HANA Smart Data Access Supported Remote Sources](#)

ABAP Platform 2025 feature scope

Data integration

DATA INTEGRATION

ABAP SQL service: ABAP-managed database procedures

Data Federation: ABAP as source

Outbound SQL via CDS external entities: Static access

SQL service, INSERT and DELETE on writable CDS view entities

Generation wizard: Outbound SQL via CDS external entities

Support for retrieving ABAP-based data via SQL service and ODBC driver using CDS SQL-based scalar functions

ABAP Platform 2025 feature scope

Process integration

PROCESS INTEGRATION

OData service consumption model

OData service preview via swagger UI

OData V4 bulk interface

OData V4 asynchronous capability with AIF integration

Application interface framework enablement

Support for assigning APIs to API packages, defining the relationship between packages and API objects

Support for creating API packages to hold various types of APIs

Support for Communication Management data via OData

ABAP Platform 2025 feature scope

Connectivity

CONNECTIVITY

Setting the AIR key as custom HTTP header field to a SOAP request

Support for direct push-based integration to enable data exchange between neighboring ABAP systems via RFC

ABAP HTTP client and server now support server-sent events (SSE)

Support for communication targets in communication arrangements

Outbound HTTP: Connectivity enablement

Outbound RFC: Connectivity enablement

ABAP Platform 2025 feature scope

Business events

BUSINESS EVENTS

Local Business Events - Final version

Filter for Outbound Bindings - Filter events before they are published to the event consumer

Support for using dynamic topics in Advanced Event Mesh - Routing and filtering outbound events, improving efficiency and targeted event delivery

Event-Driven RAP Side Effects - Define a business event to trigger a RAP side effect in your RAP business object

Business Event Logging - Capture and log business events locally within the ABAP environment

Grouping Events as Activities - Events of the same type and object occurring in the same session are grouped as activities

Success Log For Processed Events- It is now possible to configure a log for successfully processed events

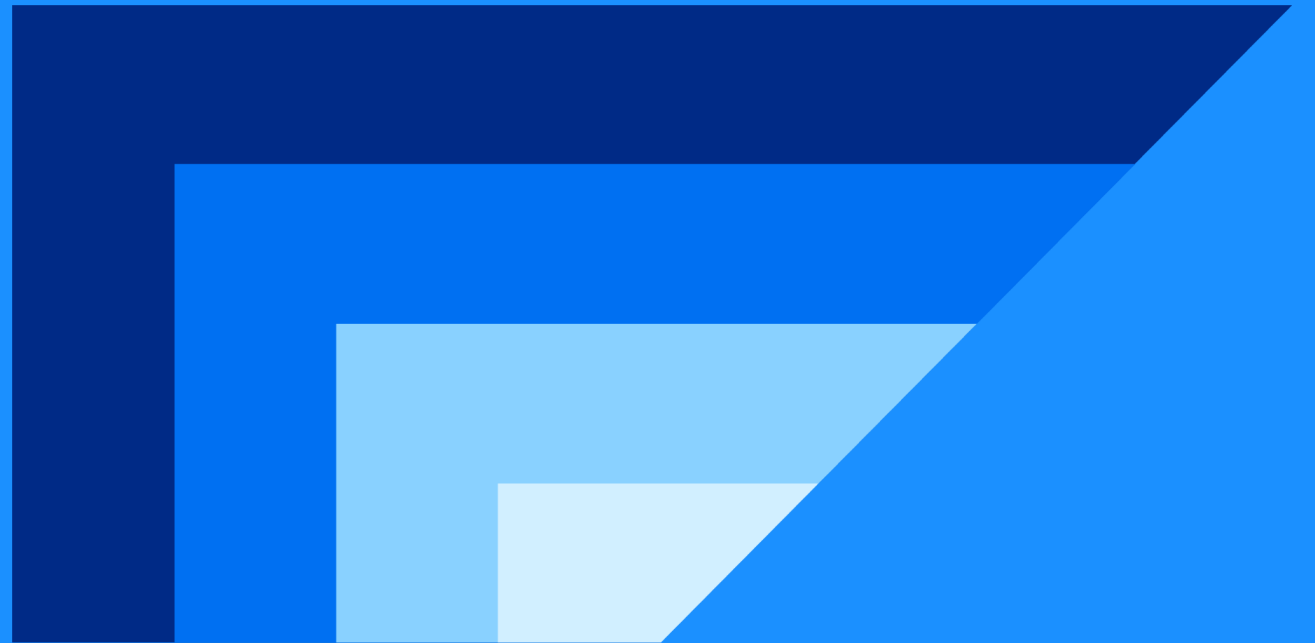
Derived event to create a new custom RAP business event based on an SAP standard event

EVENT ENABLEMENT

Support for integrating enterprise event enablement with Advanced Event Mesh service plan ([details](#))

Integrate with SAP Cloud Application Event Hub to consume the events published by SAP Order Management foundation and distribute events to your consuming applications ([details](#))

ADDITIONAL RESOURCES



Upskill yourself in Clean Core and ABAP Cloud

SAP Learning journeys valid for SAP S/4HANA Cloud and SAP S/4HANA

The screenshot shows the SAP Learning Journey interface. At the top, the SAP Learning logo is visible. The main heading is 'Practicing Clean Core Extensibility For SAP S/4HANA Cloud'. Below the heading, it indicates '5 Units', '8 hrs', 'Free until certification', and 'Certification'. The central part of the page features a video player with a 'Clean Core' diagram. To the right of the video, there is an 'Overview' section, 'Learning objectives', 'Your current experience in this topic' (with a progress bar for Beginner, Intermediate, and Advanced), 'Roles' (Developer), and 'Prerequisites' (Learning the Basics of ABAP Programming on SAP BTP, Acquiring Core ABAP Skills). At the bottom, there is a 'Start learning' button and a 'Prepare for certification' section with a QR code.

SAP Learning Journeys:

[Practicing Clean Core Extensibility for SAP S/4HANA Cloud](#)
[Acquiring Core ABAP Skills](#)

Become an

[SAP Certified Associate – ABAP Cloud Back-End Developer](#)

More SAP training and certification opportunities:
www.sap.com/education

Further information

[Extend SAP S/4HANA in the cloud and on premise with ABAP based extensions](#) | ABAP Extensibility guide

[Release Notes for ABAP Platform 2025](#) | SAP Help Portal

[Documentation for ABAP Platform 2025](#) | SAP Help Portal

[ABAP Cloud Development Guide](#) | SAP Help Portal

[ABAP RESTful Application Programming Model \(RAP\)](#) | SAP Community sub-topic page

[ABAP Roadmap Information](#) | For ABAP-based products and ABAP Cloud

[ABAP Cloud sessions at Devtoberfest in 2025](#) | [ABAP Cloud sessions at SAP TechEd 2025](#) | Blog posts (incl. replays)

Public SAP Web sites

ABAP Development Community: www.sap.com/community/topic/abap.html

ABAP Extensibility Community: <https://pages.community.sap.com/topics/abap-extensibility>

ABAP Testing and Analysis Community: <https://community.sap.com/topics/abap-testing-analysis>

SAP BTP ABAP environment Community: <https://community.sap.com/topics/btp-abap-environment>

SAP S/4HANA Cloud ABAP environment Community: <https://community.sap.com/topics/s4hana-cloud-abap-environment>

SAP products: www.sap.com/products

SAP training and certification opportunities

www.sap.com/education – e.g. trainings S4D400, S4D430, S4D437, and HA400.

<https://learning.sap.com/> – e.g., search for ABAP or ABAP Cloud

[SAP Certified Associate – ABAP Cloud Back-End Developer](#)

Thank you.

Contact information:

Olga Dolinskaja

Product Management
ABAP Platform, SAP SE
olga.dolinskaja@sap.com

Carine Tchoutouo Djomo

Product Management
ABAP Platform, SAP SE
carine.tchoutouo.djomo@sap.com